

# Gemini Robotics Data Labeling

## Hindsight Labeling: Core Guidelines

|               |                                |  |
|---------------|--------------------------------|--|
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## About this document

This document outlines the **foundational best practices** for creating high-quality annotations across **all hindsight labeling workflows**.

While these principles apply broadly, each project has unique requirements. **You must also consult the guidelines specific to your assigned workflow below.** These documents contain definitions and instructions unique to that project.

- Atari
- Omegastar
- Human

Note: If a workflow-specific instruction differs from these Core Guidelines, the workflow-specific guideline takes precedence.

## Key principles

Your primary goal is to **be specific and keep things simple**:

- You should eliminate ambiguity, not describe every visual detail.
- Add just enough information to clearly distinguish an object or location from others.
- Think about how you might tell your friend to complete an action; be that direct and simple.

## Annotation guidelines

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# 1. Define accurate segments

## 1.1 Capture meaningful actions

- **Core principle:** Each segment should capture a relatively short and intentional action. We are looking for deliberate actions, meaning actions that serve a specific purpose.
- **Duration:** For robot actions, we expect the majority of segments to fall into the 3-10 second range. For human actions, we expect the segments will be shorter, given how quickly human movements are 1-10 seconds range. These lengths are **only rules of thumb, and not hard rules**; use your best judgment.

- **Meaningful actions will likely fall into one of the following categories:**

### 1.1.1 Goal- or task-oriented actions

These are actions that successfully change the state of an object to move toward the completion of a goal or task. They are the most common operations you'll see.

Examples:

- Pick up the orange hat with your right hand.
- Restock the cracker box on the middle shelf.
- Pick up the hat and place it in the cart with your left hand.

### 1.1.2 Preparatory and supporting actions

These are actions that enable a goal or a task to be accomplished.

- **Repositioning an object:** Rotating, shifting, or otherwise adjusting an object to prepare for the next step.
  - Examples:
    - Move the tape container so it faces to the right
    - Rotate the tile with your right hand so that it is straight
    - Drop the black power cable in the center of the table (as a preparatory action to plug it in)
- **Positioning the robot or camera:** Intentionally moving the head, body, or arms to get a better view or position.
  - Example: Turn your head to the right so that you can see the apple
- **Adjusting a grip or regrasping:** Deliberate movements to refine a previous action.
  - Example: Wiggle your fingers so you only grasp one hat
    - [Video](#): :06-:16, the hand wiggles to drop a second hat that was accidentally picked up

### 1.1.3 Simultaneous and bimanual actions

Sometimes the robot or person is performing actions with each hand at the same time, in which case you can combine the actions together.

Examples:

- Open the shelf with your right hand while holding the bowl with your left hand.
- Tighten the screw with your right hand while grabbing the batch of screws with your left hand.
- Hold the dish pot handle with the left hand and use a spatula to scoop a piece of meat with the right hand. ([screenshot](#))

- Hold a plate in the left hand and use a spatula with the right hand to scoop rice from the pot and fill the top half of the plate. ([screenshot](#))

## 1.2 Capture the entire action

The segment must include the entire action, from approach to completion. The beginning of the segment should be when the action's movement starts, and the end of the segment should be when the action has fully completed.

A good rule of thumb is to start the segment when the arm leaves its resting state or changes trajectory to move toward the object. Do not wait for the hand or gripper to touch the object to start the segment; the "approach" is a critical part of the data.

### 1.2.1 "Pick and Place" actions

**Goal:** Capture the full intent of the motion. Start segments early enough to capture the "approach" trajectory, and end them when the specific motion phase is stabilized or finished.

- **Combining vs. splitting "Pick and place" actions:** Before segmenting, watch the full movement.
  - **Combined action:** If the arm picks up an object and immediately moves to place it in one fluid, continuous motion, it's usually best to keep this as one segment, i.e. pick and place in one segment.
  - **Separate actions:** If the arm pauses, stops, or changes intention significantly between the pick and place, it's likely better to split them into two segments, i.e. pick as one segment, place as a different segment.
- **"Pick" guidelines:**
  - **Where to start:**
    - Begin the segment when the arm leaves its resting state or changes trajectory toward the object (the "approach").
    - Ideally, in the very first frame of the segment, it should not be obvious which object the arm is going to pick up.
    - If the actor brings the object toward itself (retracting) rather than moving toward the destination, include that in the Pick segment.
  - **Where to end (if separate from Place):**
    - Once the object is lifted a few inches from the initial position, stable, and no longer touching the initial surface.

- **“Place” guidelines:**
  - **Where to start** (if separate from “Pick”):
    - Generally, begin the segment when the arm initiates the movement toward the placement destination.
    - As with “Pick”, it shouldn’t be immediately obvious in the first frame where the object is going.
    - It is acceptable for the start of this segment to overlap slightly with the end of the previous “Pick” segment if it helps the action feel more complete.
  - **Where to end:**
    - The Place action is complete when the object is no longer touching the actor and the arm has started moving toward the next action.
    - If the actor returns to a neutral position after placing, it is helpful to include that return motion in the segment.

### 1.3 Do not segment idle time or irrelevant motions

Do not create segments for periods where the robot or person is completely still or not engaged in meaningful action. Leave this as unsegmented time. This includes:

- Idle periods or periods of no movement at the beginning or end of an episode.
- Significant pauses occurring before or after a distinct action.

#### Notes:

- If the robot pauses *in the middle* of an action, do not treat this as idle time; this is an error.
- For any irrelevant or erratic actions

### 1.4 Do not segment unsuccessful actions

Only create segments for the successful parts of an action.

- For example, if an attempt to grasp an object fails completely, do not create a segment for it.
- Do create segments for other, successful actions within the episode.

## 2. Write accurate and unambiguous labels

The goal of these annotations is to teach the robot to understand natural human instructions. Because humans describe things in many different ways, there is rarely only one "perfect" label. Instead of looking for a single correct answer, focus on **sufficiency** and **variety**.

- **Sufficiency:** Does the label provide enough information for the robot to succeed without confusion?
- **Variety:** We require diversity in descriptions. One annotator might say "the red cup," and another might say "the cup on the left." If both distinctively identify the correct object, both are acceptable.

**Do your best to avoid repeating the same instruction, even if the actions in the video are similar.** Think of different ways an instruction might make sense to a person.

When distinguishing an object or location, use your judgment to describe the attributes that feel most natural for the scene. **Ask yourself: What's the simplest way to be sufficiently accurate?**

You have several options available to you (detailed in the following sections), such as:

- **Visual attributes** (color, shape)
- **Spatial references** (left/right, next to)
- **Specific names** (brand names, precise terminology)

### 2.1 Clearly identify objects

Your label must be specific enough to ensure the robot interacts with the intended item. However, "specific" does not mean "long." It means "unambiguous in the current context."

#### 2.1.1 Be as specific as needed to fulfill the task

Apply the "Task Failure Test" to determine how specific you need to be: **Think about if you were completing this task - what level of information would you need?**

- Scenario A (Identical/Interchangeable items):
  - **Example:** On a shelf of 10 identical soda cans, the label "pick up a soda can" is acceptable because picking any of them succeeds.
  - **Ask yourself:** If the robot picks the item next to the one you intended, would the task fail? If NO (it doesn't matter which one), a more general description is acceptable.
- Scenario B (Unique targets):

- **Example:** If one specific cup is empty and needs to be filled, "cup" is ambiguous (the robot might try to fill an already full cup). You should specify "the red cup", or "the cup on the right"
- **Ask yourself:** Should the robot pick exactly one specific object to succeed? If yes, add detail until only that object matches.

Some ways to add detail to uniquely identify the object:

- Use visual attributes: Add color, size, shape, or material.
  - the red book, the largest box, the metal bowl
- Use positional language: Describe its location relative to other objects
  - the cup to the left of the keyboard, the book on the bottom shelf

### 2.1.2 Name objects accurately

If your workflow has an official object list, use the names from that list. If your workflow does not have an official object list, consider these ways to identify the object in addition to the ways to add detail noted above:

- Use brand or packaging names if visible
  - the Campbell's soup can
  - the box of Cheerios
- Use specific common names if you're certain of them
  - cable → USB cable (only if this is accurate)

### 2.1.3 Always describe objects as real

Describe what you see, not what the object represents or is used for.

- Do not refer to items as "fake", "mock", "toy", or "model".
  - toy car → small red car
  - fake plant → plastic plant in a pot
- Do not refer to "robot" parts
  - robot hands → hands
  - robotic arms → arms

### 2.1.4 Use Standard American English names for objects

Refer to objects with their Standard American English names. Avoid regional terms.

- trolley → shopping cart

## 2.2 Describe the action

### 2.2.1 Use descriptive verbs

Use verbs that are clear and not likely to be misinterpreted. Usage of general-purpose verbs (like put, get, take, or move) usually means there are also more specific ways of describing

the action. When possible, use a more descriptive alternative that most closely describes the method of doing something.

- Ask yourself, “**What movement is actually happening?**” and describe that.
  - Ambiguous: Adjust the thermostat
  - What’s actually happening: The hand is turning a dial.
  - Descriptive: Rotate the thermostat dial clockwise to 70 degrees
- Because we want the robot to understand a variety of instructions, we are looking for a diversity of verbs.
  - **Do your best to avoid using the same verb over and over**, even if the actions in the video are the same or similar.
  - Do not use the word adjust.
  - Think of alternatives that accurately describe the action, which may look like:

| More general                                                                                                         | More descriptive                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Put the cable in the slot<br>Put the key into the lock<br>Put the lid on the jar                                     | Plug / insert the cable into the slot<br>Insert the key into the lock<br>Screw / snap the lid onto the jar            |
| Move the chair under the desk<br>Move the cart forward<br>Move the box to the left<br>Move the pitcher to pour water | Push the chair under the desk<br>Pull the cart forward<br>Slide the box to the left<br>Tilt the pitcher to pour water |
| Pick the apple from the bowl<br>Pick the tray from the oven                                                          | Take the apple from the bowl<br>Remove the tray from the oven                                                         |

### 2.2.2 Always specify the hand used

**[MANDATORY]** Always identify which hand(s) performs the action, even if it seems obvious. Use this for both single-handed and bi-manual (using two hands) actions.

| ✗ Vague (avoid)   | ✓ Descriptive (use)                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------|
| Pick up the apple | Pick up the apple <b>with the right hand</b>                                                             |
| Lift the tray     | <b>With both hands</b> , lift the tray onto the rack                                                     |
| Thread the needle | Hold the needle in the left hand, and with the right hand, pass the thread through the eye of the needle |



## 2.3 Clearly describe locations

Specify the location information necessary for the robot to complete the interaction successfully. You do not always need both a starting and ending location if one is obvious from context.

### 2.3.1 Be as specific as needed to fulfill the task

Apply the "Task Failure Test" to determine how specific you need to be regarding locations: **Think about if you were completing this task—what level of location information would you need?**

- Scenario A (General locations acceptable):
  - **Example:** Placing an apple into a large, empty fruit bowl. The label "place the apple in the white bowl" is acceptable because placing it anywhere inside the bowl succeeds.
  - **Ask yourself:** If the robot places the item slightly to the left or right of where I intended, would the task fail?
  - If **NO** (it doesn't matter exactly where it lands, as long as it is in the general area), a more general description is acceptable.
- Scenario B (Precise locations required):
  - **Example:** Inserting a book into a specific gap on a crowded shelf. "Place the book on the shelf" is ambiguous (the robot might try to jam it where there is no space).
  - You should specify "place the book between the red book and the blue vase".
  - **Ask yourself:** Must the robot reach one exact spot to succeed?
  - If **YES**, add detail until only that spot matches.

### 2.3.2 Use relative referencing

When an object's location isn't immediately obvious, anchor it to stable landmarks.

Your goal is to **isolate** the object, not to fully describe its environment. Use as many reference points as you need to be accurate, but no more.

Ask yourself: *What is the least amount of detail that separates this object from its neighbors?*

Prioritize static objects as reference points

Refer to permanent fixtures like shelves, walls, or other stationary objects.

-  Good: Pick up the white tube that's on the shelf beneath the orange book.
-  Good: Grab the book to the right of the candle.

Use stable references from the robot's perspective

When referencing the robot in the video, choose terms that won't change as it moves.

"Closest to" is often a very stable reference.

-  Good: Take out the white tube that's **closest to your right hand**.
-  Bad: While holding the object in your right hand - Put the object **to the right of your arm**. (The arm's position will change, making the reference point useless.)

Avoid absolute or complex coordinates

Do not use specific coordinates unless they are part of a pre-defined grid system for the scene. Relative language is almost always better.

-  Bad: Grasp a white tube on the left blue tray **placed on the bottom third shelf**.
-  Good: Grasp the white tube on the blue tray **on the shelf beneath the vase**.

## 2.4 Define the extent & goal of the motion

Applies to actions where the robot needs to know **when to stop**, or **how to place something**.

**Remember:** Imagine that you are telling someone who doesn't know what the task is how to do it. They would likely need to have all of this same information!

### 2.4.1 Moving the actor (person or robot)

This applies to actions like turning, bending, looking, etc. If you tell the actor to "turn left", it might spin forever. You must specify the **cue** that tells the actor to stop.

**Remember:** The object you intend to interact with must be visible (or become visible) during the segment. If the actor simply moves its body or head, it's usually trying to see or find something. You must specify what the visual target is.

|  Vague (avoid) |  Descriptive (use) |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Turn to the left                                                                                  | Turn to the left <b>until you can see black belt</b> .                                                |
| Bend down                                                                                         | Bend down <b>until the bottom shelf is visible</b> .                                                  |
| Look up                                                                                           | Look up <b>until the top of the cabinet is in view</b> .                                              |

## 2.4.2 Adjusting an object

This applies to actions like pushing, sliding, pulling, rotating, folding, etc.

When moving an object, you must describe the final state of the object. A vague goal like “to create space” isn’t enough -- the robot needs to know how much space or where the object should end up.

- After describing an object movement, ask yourself if you know *why* it occurred. This reveals the action's purpose.
  - Note: The "Use descriptive verbs" guideline taught you to replace `adjust` with a verb like `slide`. This guideline teaches you to explain the reason for that slide.
- Do not use the word `adjust`.

|  Vague (avoid) |  Descriptive (use)                                            |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Push the chips backward                                                                         | Push the chips backward <b>until the bag is aligned with the other chips.</b>                                                                  |
| Slide the box to create space at the front                                                      | Slide the box <b>until it's in the back half of the shelf</b><br><br>Slide the box <b>so we have space for three boxes from the front edge</b> |
| Fold the flap                                                                                   | Fold the flap <b>down to close the box</b>                                                                                                     |
| Rotate the bottle                                                                               | Rotate the bottle <b>so the label faces forward</b>                                                                                            |
| Nudge the box to the right                                                                      | Nudge the box to the right <b>to align it with the edge of the table</b>                                                                       |

## 2.4.3 Define the final object orientation

Use this for actions where the landing or final state is critical. When placing an object, the final position is often just as important as the location. If the object needs to face a specific way to be useful, you must describe it.

|  Vague (avoid) |  Descriptive (use) |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Place the book on the shelf                                                                       | Place the book on the shelf <b>with the spine facing out</b>                                          |
| Insert the battery                                                                                | Insert the battery <b>with the positive (+)</b>                                                       |

|  |                |
|--|----------------|
|  | side facing up |
|--|----------------|

### 3. Write simple and easy-to-understand labels

Your labels are direct instructions for a robot. They should be written like commands in a recipe: simple, clear, and focused on a single action.

See [Hindsight Labeling Rewrite Examples](#) for how to approach simplifying labels.

#### 3.1 Write as a direct command

Start your sentence with a verb. You are telling the robot what to do, not describing what someone else is doing.

| ✗ Narrative (avoid)                         | ✓ Direct command (use)           |
|---------------------------------------------|----------------------------------|
| The person is picking up the red block.     | Pick up the red block.           |
| Next, the blue cup was placed on the table. | Place the blue cup on the table. |
| What he does now is open the drawer.        | Open the top drawer.             |

#### 3.2 Include only necessary details

The goal is to be unambiguous, not overly descriptive. Add just enough detail to make the command clear and then stop. Omit filler words and unnecessary adjectives.

- **The best way to check for clarity is to read your label out loud.** If it sounds unnatural, confusing, or is hard to say in one breath, it's too complex and needs to be simplified.
- Before finalizing your label, ask yourself: **"If I gave this instruction to a person, could they perform the action correctly without asking for clarification?"** This is your final quality check to ensure your command is truly clear and actionable.

### 4. Handle edge cases and errors

While it's important to capture all meaningful actions, it is equally important to omit or flag movements that are not intentional. This ensures the training data is clean and focused on correct behaviors.

## Category: Motion Quality

### 1. Collision

- Guideline:
  - The robot and its parts must not unintentionally collide with the environment
  - Robot knocks over the target object or other crucial objects

### 2. Erratic motion

- Guideline:
  - Non-optimal in terms of directory
  - Robot has out of controlled motions, non-ideal motion
  - When robot overshoots object vs going directly to the object

### 3. Paused motion

- Guideline:
  - Only use paused motion if there are pauses while the robot is performing a task.
    1. Robot pauses for > 1 sec
    2. Optimal trajectory but stops on the path to the object
    3. Robot motion hesitates
  - Do not use paused motion if the the robot is not moving at the very the [start](#) or [end](#) of the video as it is expected pause motion may happen

### 4. Completion adherence

- Guideline: Flag as an error if the video shows no completed or successful action
  - In these cases, create segment highlighting entire video
  - [Example](#)

## Grasp & Manipulation

### 5. Firm grasp

- Guideline:
  - The robot must establish a secure grasp on the object.
  - The object should not be held precariously or be at risk of being easily dropped.
  - Flag as an error if the grasp is not firm or if the object is dropped more than once use `multiple_attempts`.

### 6. Multiple attempts

- Guideline:
  - Excessive Drops: Flag as error if object is dropped more than two times occur without successful recovery. Do not flag if an object is dropped and successfully recovered within two attempts.

- Unnecessary Transfers: The robot repeatedly transfers objects between grippers/hands without a clear purpose.

Category: Visual/Perception

### 7. Camera

- Guideline: Camera feeds should be clear and free from issues like blurriness, tinting, corruption, stuttering, or freezing. Flag as an error if image quality is compromised.

### 8. Field of view

- Guideline: The robot's arms, grippers, and the primary object(s) being manipulated must be within the field of view of at least one camera.
  - Ex: If the object is only viewable from the poc camera but not the overview camera's (example)

### 9. Other

- Recommendation: Any unforeseen blatant errors not covered by other categories. This should be used sparingly.

Category: Human data labeling

### 10. Bad tracking

## Appendix

### Key definitions

- **Episode:** The entire video recording from start to finish.
  - e.g., *A 2-minute video showing robotic arms tidying a space*
- **Scene:** The physical environment and setup where the activity happens.
  - e.g., *A shelf with books and various objects on different shelves and on the floor*
- **Segment:** A specific portion of the Episode with a defined start and end time, created to isolate a particular Action.
  - e.g., *A 5-second clip from 0:32 to 0:37 showing a book being put on a shelf*
- **Label (aka Instruction):** A written command that describes the Action within a Segment. It tells the robot what to do.
  - e.g., *"Place the blue book on the shelf beneath the vase"*
- **Annotation:** The final data output created by an annotator, which combines a video Segment with a written Label.

## Action hierarchy

|                                                                        |                                                                                                              |                                                                                                                                                                                                         |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Level 1<br><b>Goal</b>                                                 | The high-level purpose or desired final outcome.                                                             | <ul style="list-style-type: none"> <li>Clean the room</li> </ul>                                                                                                                                        |
| Level 2<br><b>Task</b>                                                 | A major objective that must be completed to achieve the Goal.                                                | <ul style="list-style-type: none"> <li>Put all the books on the book shelf</li> </ul>                                                                                                                   |
| Level 3<br><b>Step</b><br>(aka Natural instruction for Omegastar)      | A single, direct command to make progress on a Task.                                                         | <ul style="list-style-type: none"> <li>Place the blue book on the shelf</li> </ul>                                                                                                                      |
| Level 4<br><b>Specification</b>                                        | A precise command that clarifies how a Step is performed, including its specific method or required outcome. | <ul style="list-style-type: none"> <li>Place the blue book on the shelf with the cover facing you</li> </ul>                                                                                            |
| Level 5<br><b>Movement</b><br>(aka Detailed instruction for Omegastar) | A description of a simple physical movement required to execute a Step or Specification.                     | <ul style="list-style-type: none"> <li>Grab the blue book with the left arm</li> <li>Lift the book</li> <li>Rotate the book on the shelf with your left hand so that the cover is facing you</li> </ul> |